

QuantumLab

Mathematical Foundations

A rigorous reference for the mathematics implemented in the QuantumLab open-source quantum simulator.
Version 2.0.0.

QuantumLab Mathematical Foundations

Chapter 1: Linear Algebra Foundations

Complex vectors

An n -qubit pure state is a unit vector $|\psi\rangle \in \mathbb{C}^{2^n}$. QuantumLab stores amplitudes as `numpy.complex128` arrays of shape $(2^n,)$.

Inner product

$\langle \phi | \psi \rangle = \sum_i \phi_i^* \psi_i$. Normalization requires $\langle \psi | \psi \rangle = 1$ (tolerance $1e-9$).

Outer product

Tensor product

For single-qubit gates U on wire q of an n -qubit register:

$U_{\text{full}} = G \otimes \dots \otimes G \otimes I$ where $G_i = U$ if $i = q$, else I .

QuantumLab builds this via `numpy.kron` in `_operators.py`.

Unitary matrices

A gate U satisfies $U^\dagger U = I$. Validated by `assert_valid_gate` with tolerance $1e-10$.

Chapter 2: Qubits and Measurement

Computational basis

Big-endian convention: $q[0]$ is the most significant bit.

$|00\rangle$ index 0, $|01\rangle$ index 1, $|10\rangle$ index 2, $|11\rangle$ index 3

Bloch sphere

Single-qubit pure states map to (θ, ϕ) via reduced density matrix:

$\rho = (I + xX + yY + zZ)/2$, $(x,y,z) = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$.

Born rule

Measurement probability of basis state $|i\rangle$ is $p_i = |a_i|^2$ for pure states, or $p_i = \rho_{ii}$ for density matrices.

Measurement in QuantumLab

The `M` gate is a visualization marker - it does not collapse the state. Probabilities are always computed from the full post-circuit state.

Chapter 3: Gate Mathematics

Pauli gates

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Hadamard

$$H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Maps $|0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$.

Rotations (Qiskit convention)

$$RX(\theta) = \exp(-i\theta/2 X) \quad RY(\theta) = \exp(-i\theta/2 Y) \quad RZ(\theta) = \exp(-i\theta/2 Z)$$

CNOT

On $|c,t\rangle$: $CNOT|c,t\rangle = |c, c\oplus t\rangle$.

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Chapter 4: Tensor Products and Embedding**Single-qubit embedding**

Gate H on qubit 0 of 2-qubit system:

$$H_{full} = H \otimes I \quad (\text{big-endian: } q[0] \text{ is left factor})$$

Gate H on qubit 1:

$$H_{full} = I \otimes H$$

Non-adjacent two-qubit gates

A 4×4 gate on wires (q_0, q_1) is embedded via SWAP permutation:

$$U_{full} = P \cdot (U \otimes I \otimes I \otimes I) \cdot P$$

where P routes wires q_0, q_1 to positions 0, 1.

Chapter 5: Controlled Gates**Projector formalism**

$$C_U = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$$

CNOT truth table**Bell state construction**

H on qubit 0, then CNOT(0,1):

$$|00\rangle \rightarrow (|00\rangle + |10\rangle)/\sqrt{2} \rightarrow (|00\rangle + |11\rangle)/\sqrt{2} = |\Phi^+\rangle$$

Chapter 6: Density Matrix Formalism

Definition

$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ with $\text{Tr}(\rho) = 1$, $\rho = \rho^\dagger$, $\rho \geq 0$.

Evolution

$\rho' = U \rho U^\dagger$ for unitary U .

Partial trace

$\rho_A = \text{Tr}_B(\rho_{AB})$. QuantumLab traces out qubits not in keep_qubits (sorted ascending).

Purity

$P = \text{Tr}(\rho^2)$. $P = 1$ for pure states; $P < 1$ for mixed states.

Entanglement entropy

$S(\rho_A) = -\text{Tr}(\rho_A \log \rho_A)$ in bits. Bell state: $S = 1$.

Chapter 7: Noise and Kraus Operators

CPTP maps

$\rho' = \sum_i K_i \rho K_i^\dagger$ with $\sum_i K_i^\dagger K_i = I$.

Amplitude damping (T1-like)

$K_0 = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{bmatrix}$, $K_1 = \begin{bmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{bmatrix}$

Phase damping (T2-like)

$K_0 = \text{diag}(1, \sqrt{1-\gamma})$, $K_1 = \text{diag}(0, \sqrt{\gamma})$

Off-diagonals scale by $\sqrt{1-\gamma}$. For Ramsey T2: $\gamma = 1 - \exp(-2t/T_2)$.

Depolarizing

$K_0 = \sqrt{1-p} I$, $K_1 = \sqrt{p/3} X$, $K_2 = \sqrt{p/3} Y$, $K_3 = \sqrt{p/3} Z$

Chapter 8: Quantum Algorithms

Bell state

Circuit: $H(0)$, $\text{CNOT}(0,1)$. Probabilities: $[0.5, 0, 0, 0.5]$.

GHZ (3 qubit)

$H(0)$, $\text{CNOT}(0,1)$, $\text{CNOT}(1,2)$. Probabilities: 0.5 at $|000\rangle$ and $|111\rangle$.

Deutsch algorithm

Determines if $f:\{0,1\} \rightarrow \{0,1\}$ is constant or balanced in one query.

Deutsch-Jozsa

Generalizes to n -bit functions; constant functions yield $|0^n\rangle$.

Bernstein-Vazirani

Recovers hidden bit string s from $f(x) = s \cdot x \bmod 2$.

Grover search

Amplitude amplification; one iteration on 2 qubits boosts marked state probability.

Quantum Fourier Transform

Basis change $|j\rangle \rightarrow (1/\sqrt{N}) \sum_k e^{2\pi i jk/N} |k\rangle$. Built from H and controlled RZ gates.

Chapter 9: Verification Against Qiskit**Methodology**

1. Build identical circuit in QuantumLab and Qiskit
2. Apply wire remap: $q_{\text{qiskit}} = n - 1 - q_{\text{qlab}}$ (endianness alignment)
3. Compare statevectors with $\text{atol} = 1e-10$ (global phase ignored)
4. Compare probabilities, fidelity, and entropy independently

Test suite

150 pytest tests in `quantum-simulator-core/tests/` including:

- * `test_math_validation.py` : unitarity, dimensions, Born rule, Kraus completeness
- * `test_qiskit_reference.py` : algorithm cross-validation

Bug fixed in this sprint

T2 phase damping: corrected gamma mapping so coherence decays as $\exp(-t/T_2)$, not $\exp(-t/(2T_2))$.

Running validation

```
pip install -e "quantum-simulator-core[validation]"
pytest quantum-simulator-core/tests/ -v
python quantum-simulator-core/scripts/generate_math_pdf.py
```

QuantumLab v2.0.0 : MIT License. This document accompanies the mathematical audit report.